Deep Learning and Optimization

Unpacking Transformers, LLMs and Diffusion

Session 6

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Summary of Session 5

Convnets as inductive bias for computer vision tasks.

Convnets are very expensive from a compute/memory perspective (1x1 trick)

More depth does not mean better performance.

ResNets and bottlenecks unlock performance and efficiency.

	Session	Date	Content
Foundations	1	Jan, 28	Intro to DL TP: micrograd
	2	Feb, 4	Fundamentals I: inductive bias, loss functions TP: bigram, MLP for next character prediction
	3	Feb, 11	Fundamentals II: DL architectures TP: tensor-based models
Applications	4	Feb, 18	Attention & Transformers TP: GPT from scratch
	5	Feb, 25	DL for Computer vision TP: convnets on CIFAR-10
	6	Mar, 11	VAE and Diffusion TP: diffusion from scratch Quiz / Exam

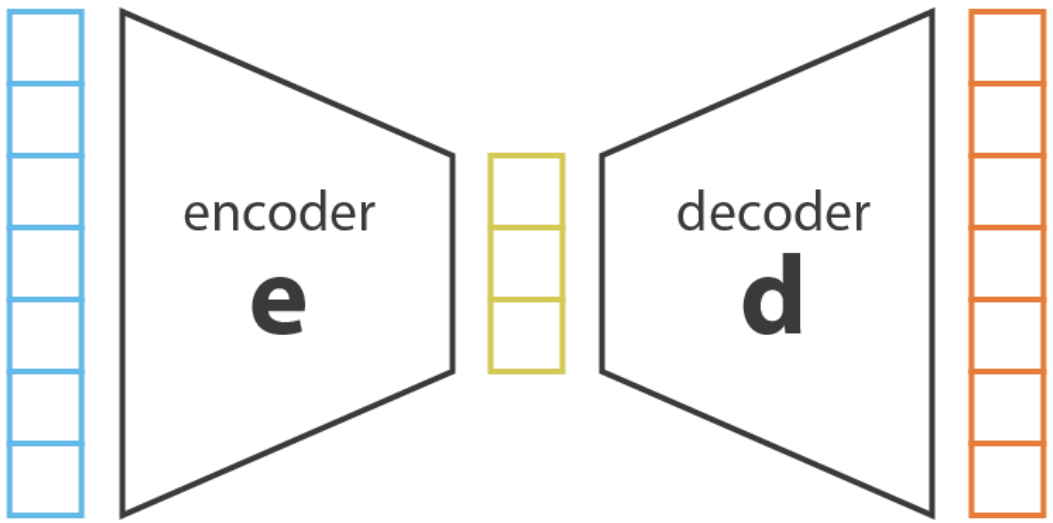
Agenda for today

1. VAEs
2. Diffusion

Variational Auto-Encoders (VAE)

First introduced in "Auto-Encoding Variational Bayes", Diederik P. Kingma and Max Welling, 2014

Auto-encoders



x

initial data
in space R^n

e(x)

encoded data
in latent space R^m (with m<n)

d(e(x))

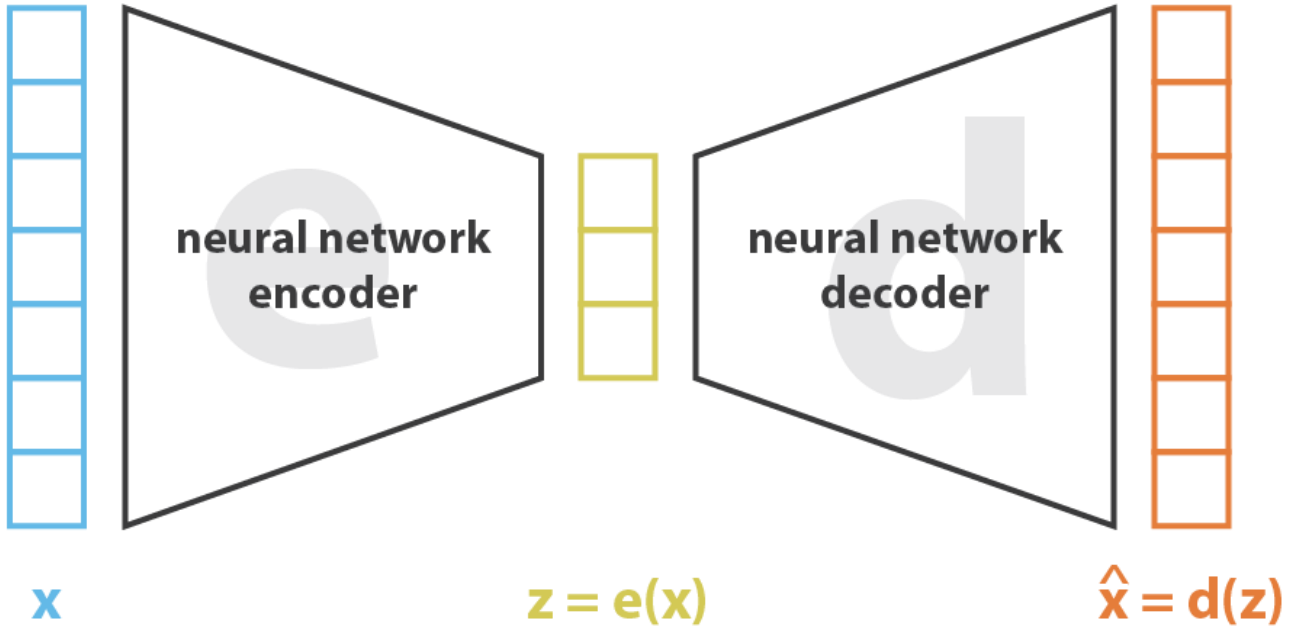
encoded-decoded data
back in the initial space R^n

$x = d(e(x))$ ➡ **lossless encoding**
no information is lost when reducing the number of dimensions

$x \neq d(e(x))$ ➡ **lossy encoding**
some information is lost when reducing the number of dimensions and can't be recovered later

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

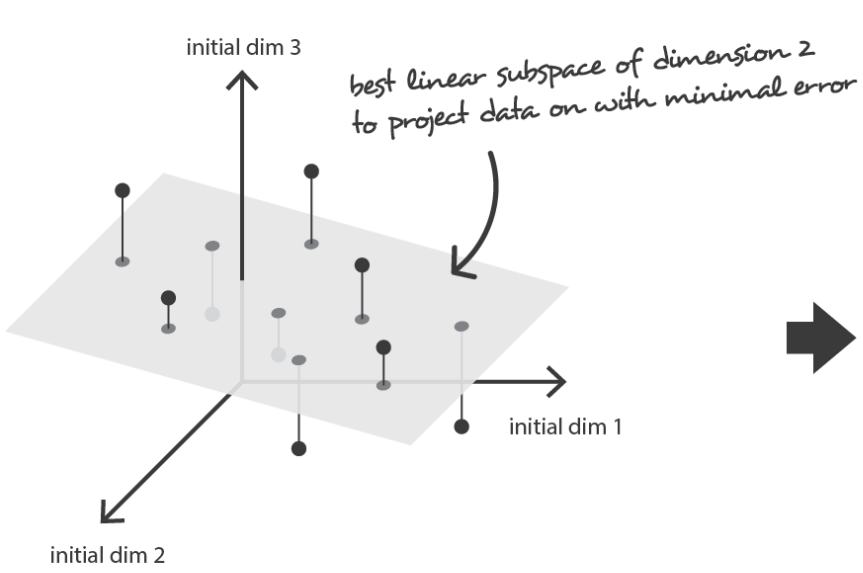
Auto-encoders



$$\text{loss} = \| x - \hat{x} \|^2 = \| x - d(z) \|^2 = \| x - d(e(x)) \|^2$$

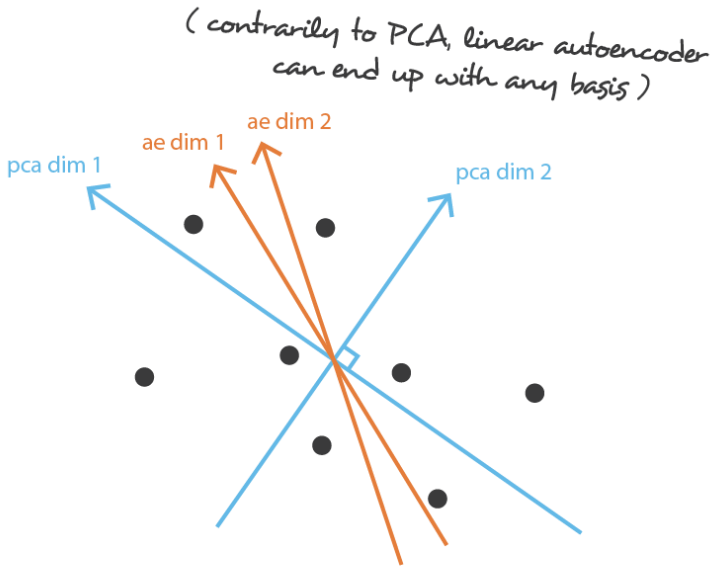
Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

Auto-encoders



Data in the full initial space

In order to reduce dimensionality, PCA and linear autoencoder target, in theory, the same optimal subspace to project data on...

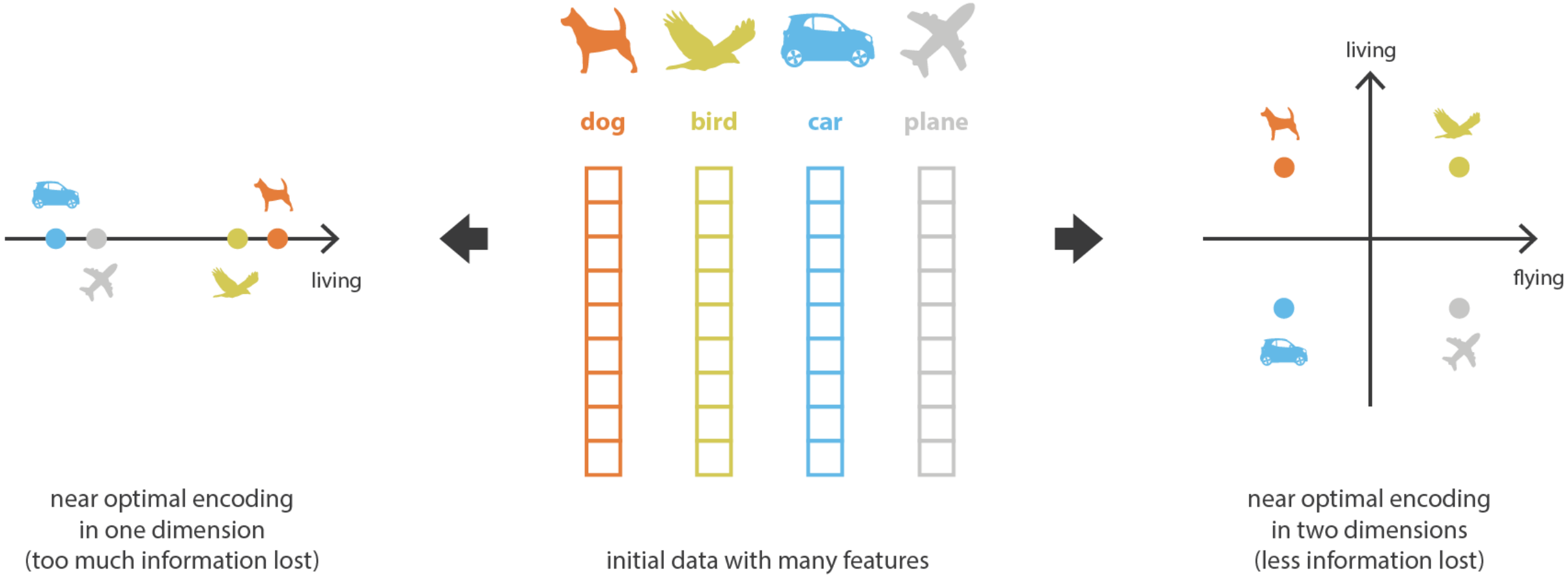


Data projected on the best linear subspace

... but not necessarily with the same basis due to different constraints (in PCA the first component is the one that explains the maximum of variance and components are orthogonal)

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

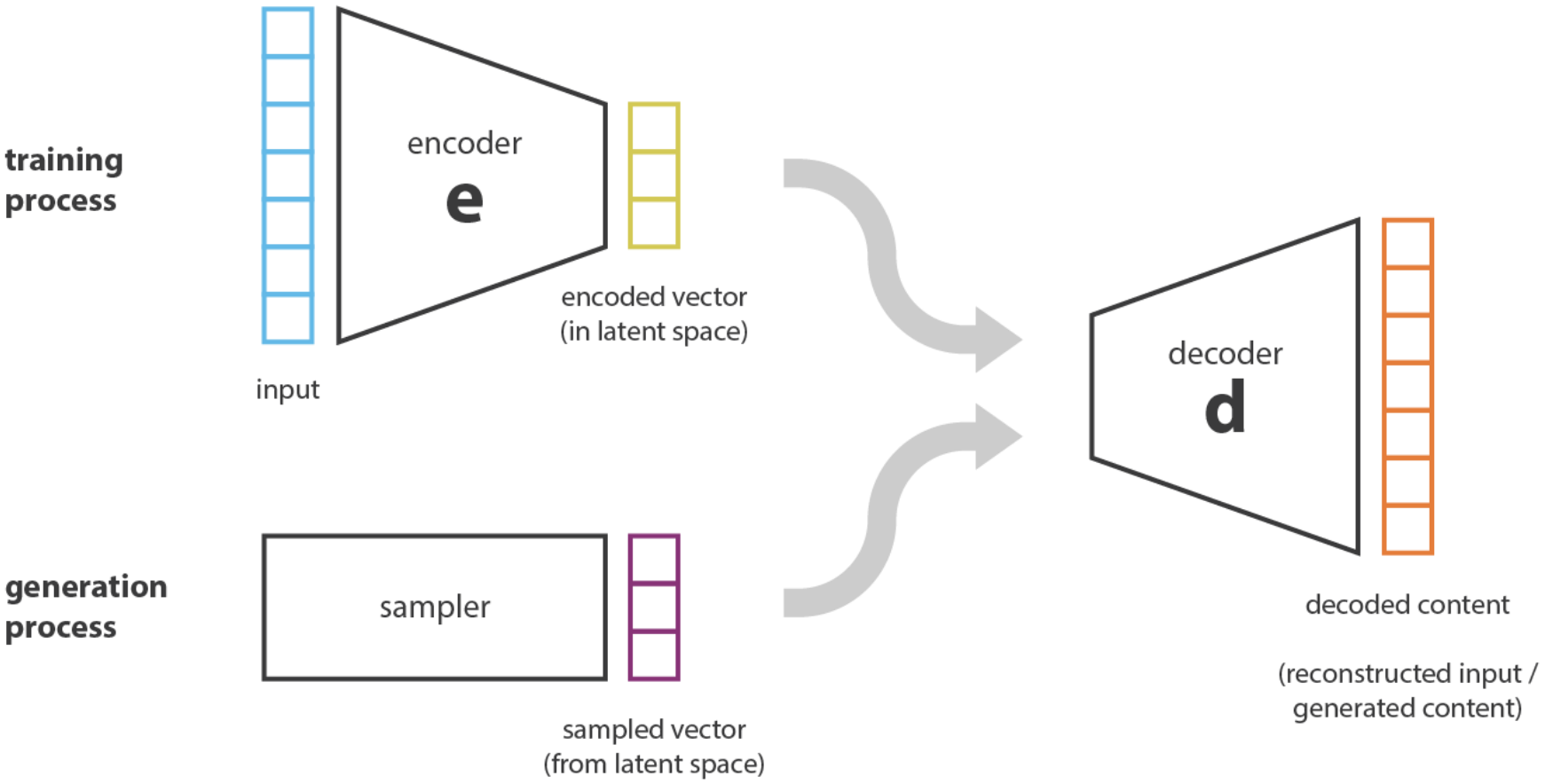
Auto-encoders



When reducing dimensionality, we want to keep the main structure there exists among the data.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

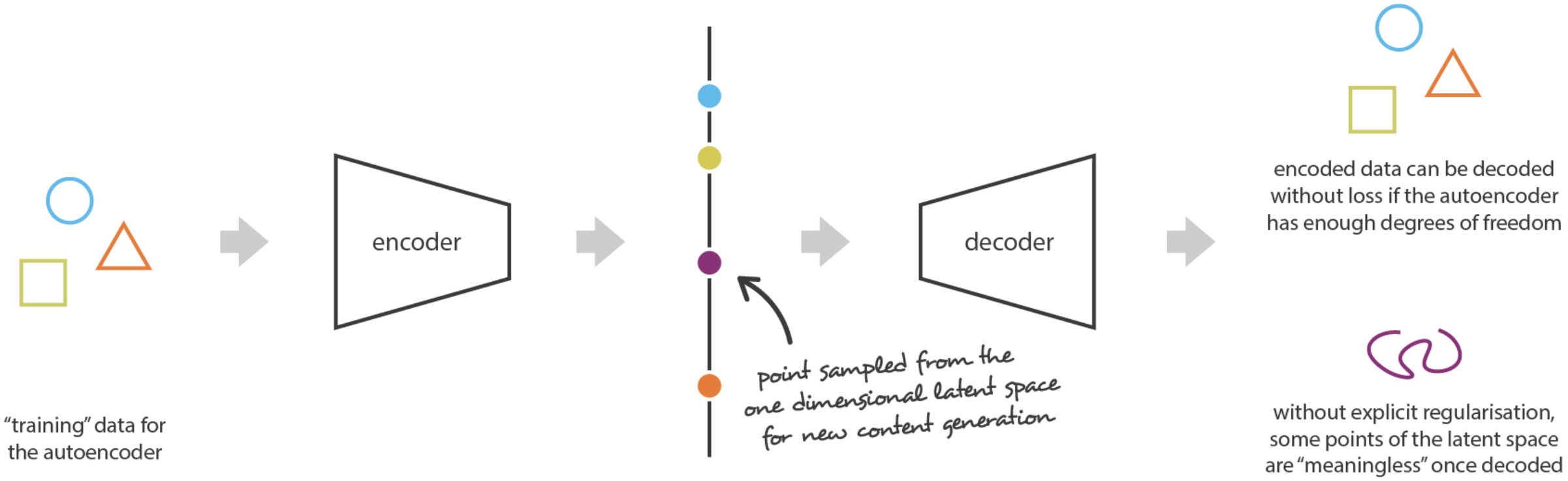
Auto-encoders



In theory, auto-encoders can be used for data generation.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

Auto-encoders



Irregular latent space prevent us from using autoencoder for new content generation.

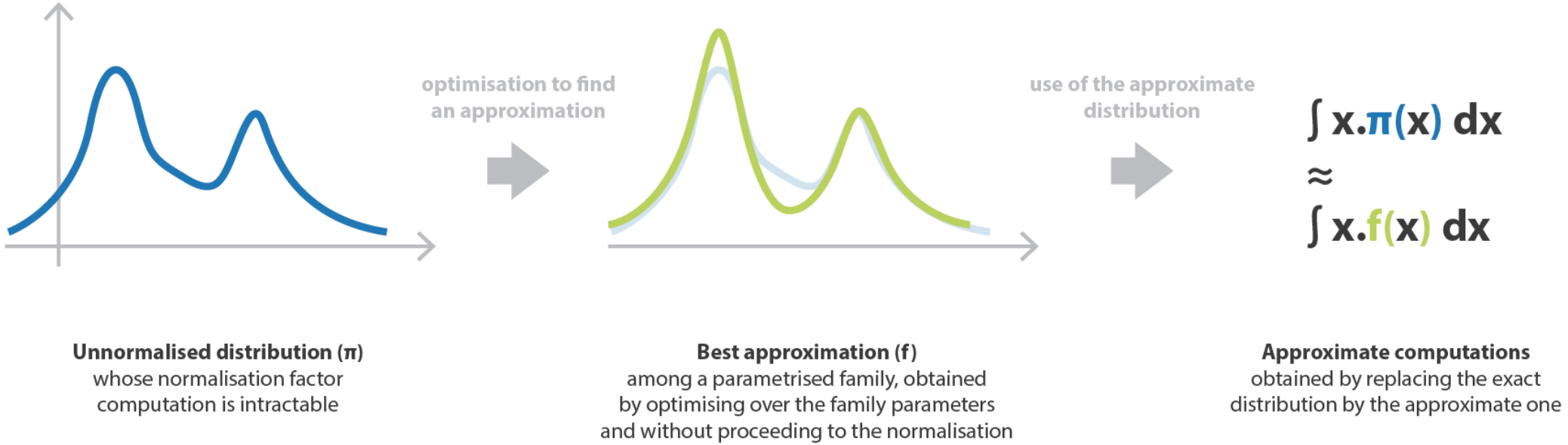
Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

Variational auto-encoders

A variational autoencoder can be defined as being an autoencoder whose training is regularised to avoid overfitting and ensure that the latent space has good properties that enable generative process.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

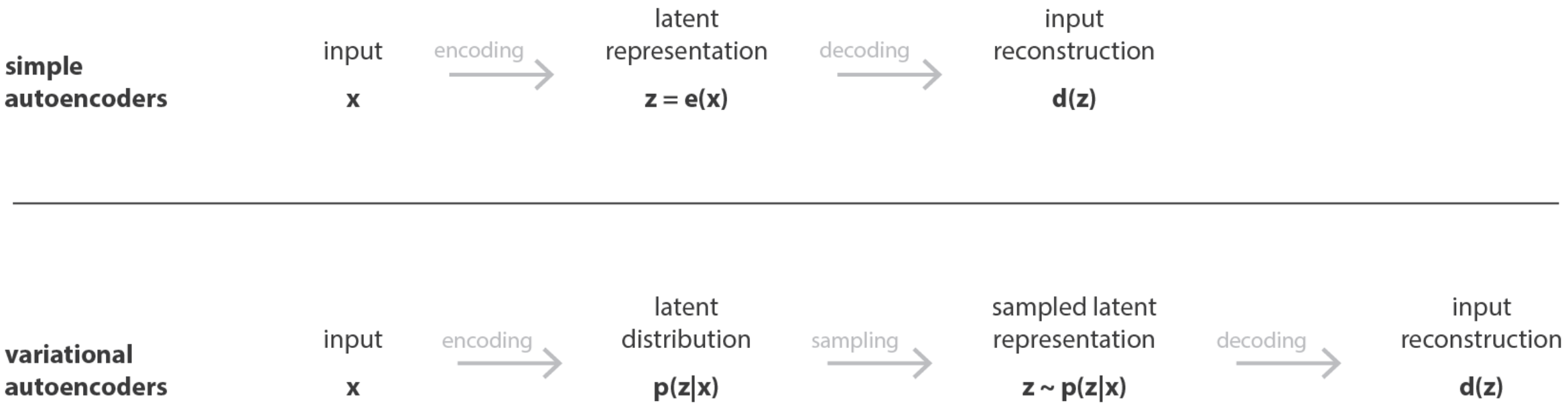
Variational inference



Variational Inference methods that consist in finding the best approximation of a distribution among a parameterised family.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

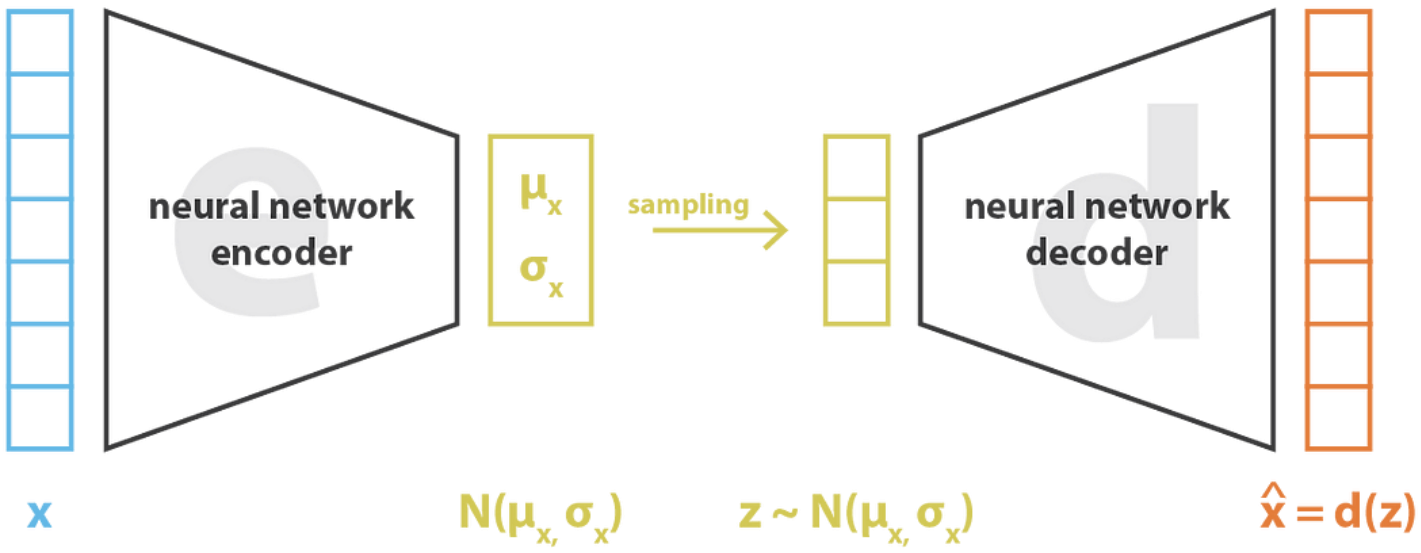
Variational Auto-encoders (VAEs)



Difference between autoencoder (deterministic) and variational autoencoder (probabilistic).

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

Variational Auto-encoders (VAEs)

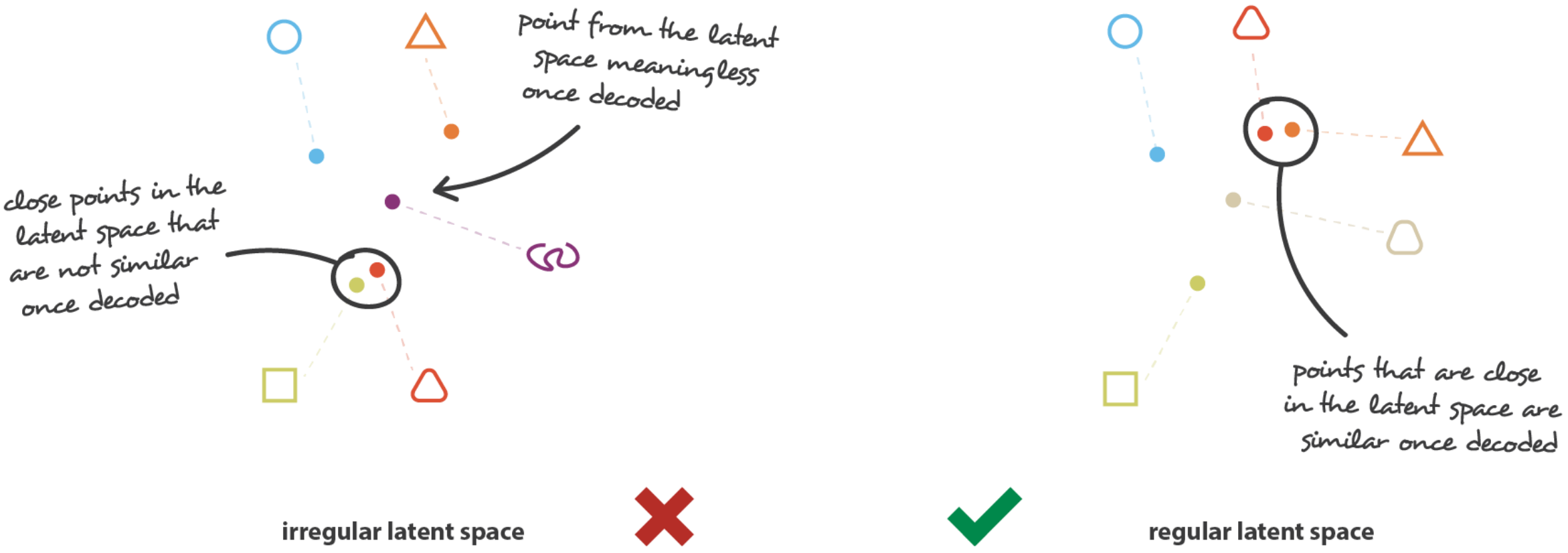


$$\text{loss} = \| x - \hat{x} \|^2 + KL[\, N(\mu_x, \sigma_x), N(0, I)\,] = \| x - d(z) \|^2 + KL[\, N(\mu_x, \sigma_x), N(0, I)\,]$$

In variational autoencoders, the loss function is composed of a **reconstruction term** (that makes the encoding-decoding scheme efficient) and a **regularisation term** (that makes the latent space regular).

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

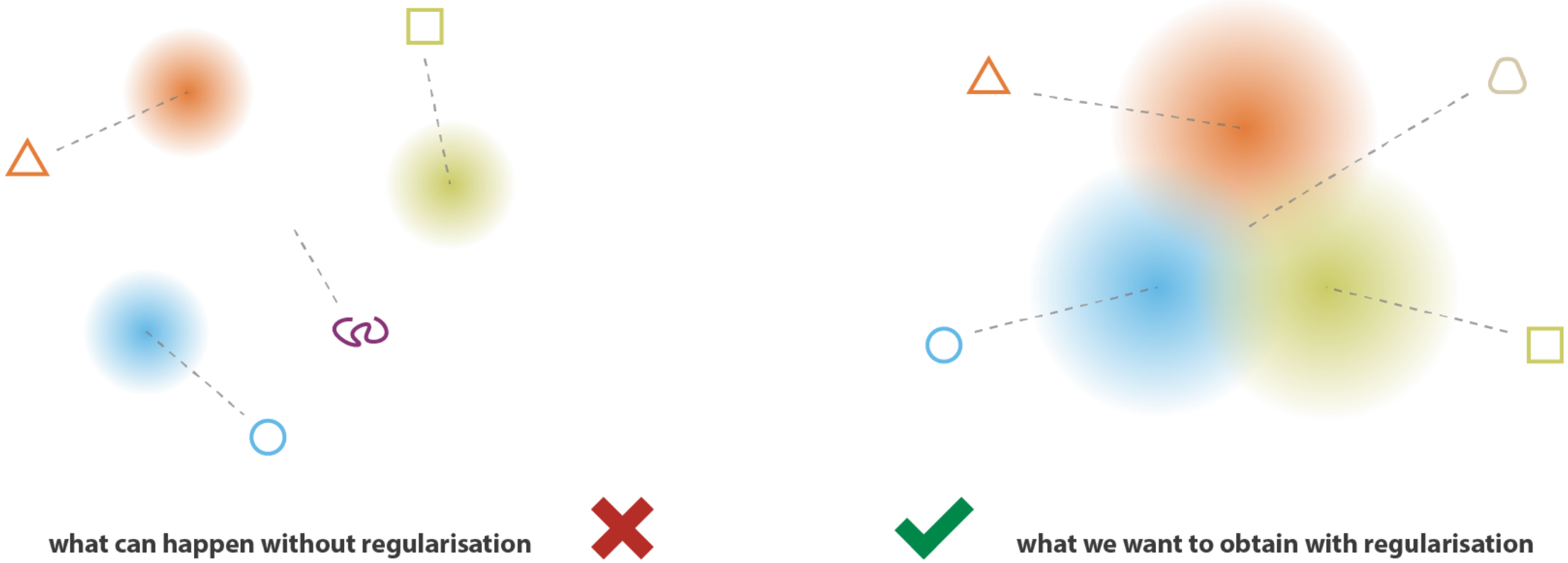
Variational Auto-encoders (VAEs)



Difference between a “regular” and an “irregular” latent space.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

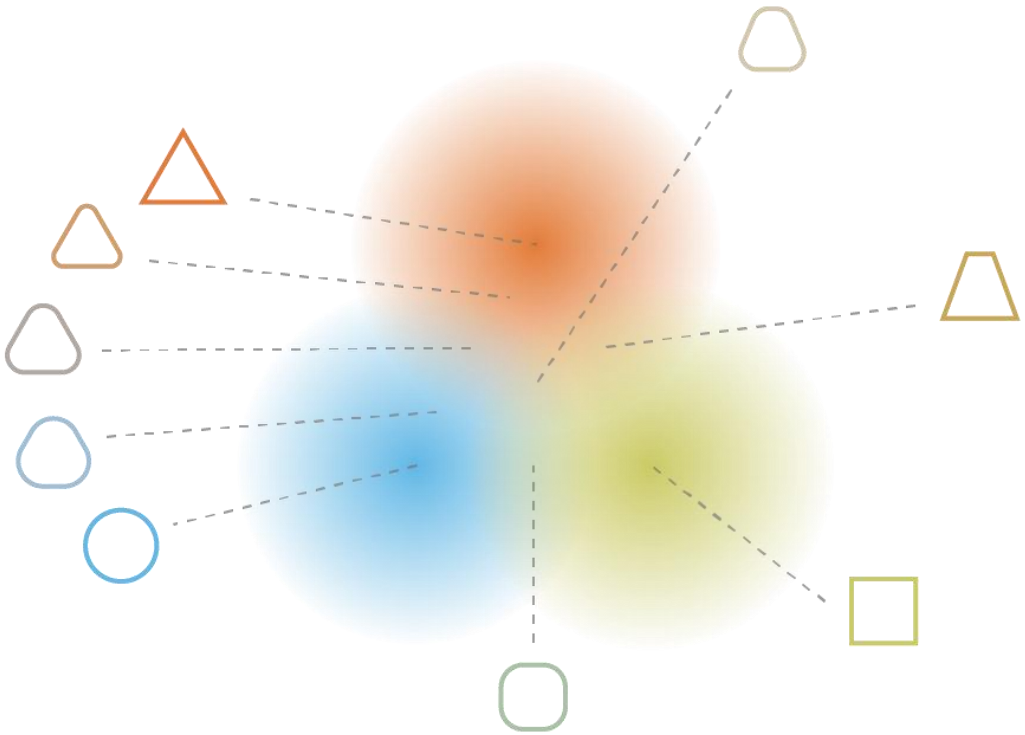
Variational Auto-encoders (VAEs)



The returned distributions of VAEs have to be regularised to obtain a latent space with good properties.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

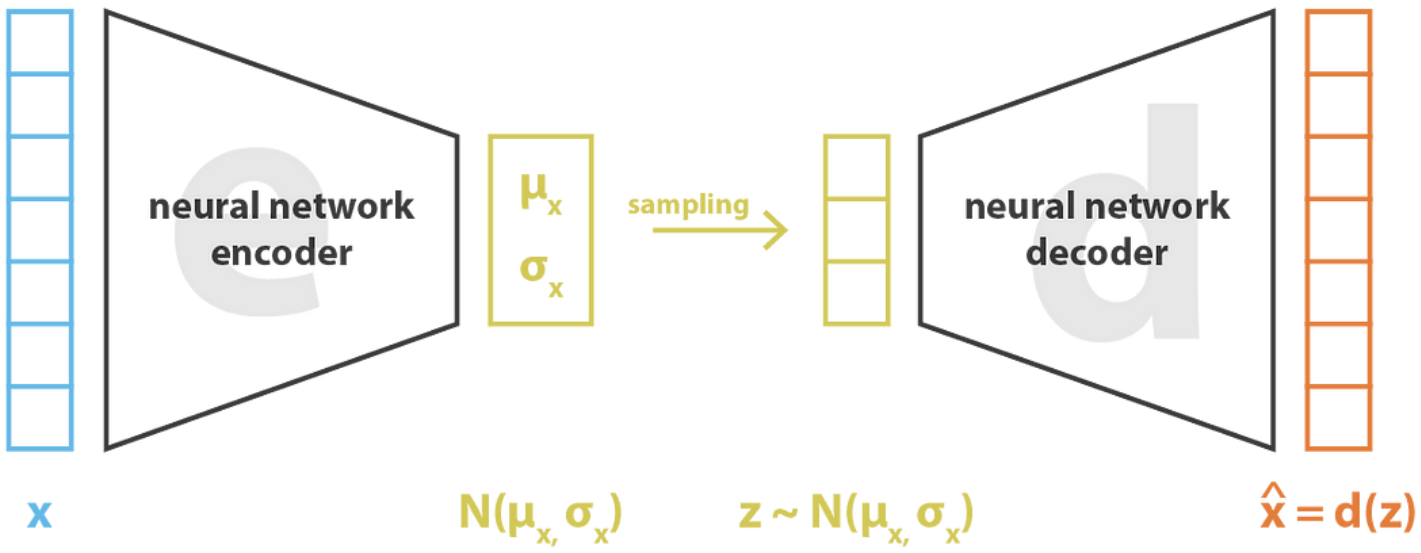
Variational Auto-encoders (VAEs)



Regularisation tends to create a “gradient” over the information encoded in the latent space.

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

Variational Auto-encoders (VAEs)



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In variational autoencoders, the loss function is composed of a **reconstruction term** (that makes the encoding-decoding scheme efficient) and a **regularisation term** (that makes the latent space regular).

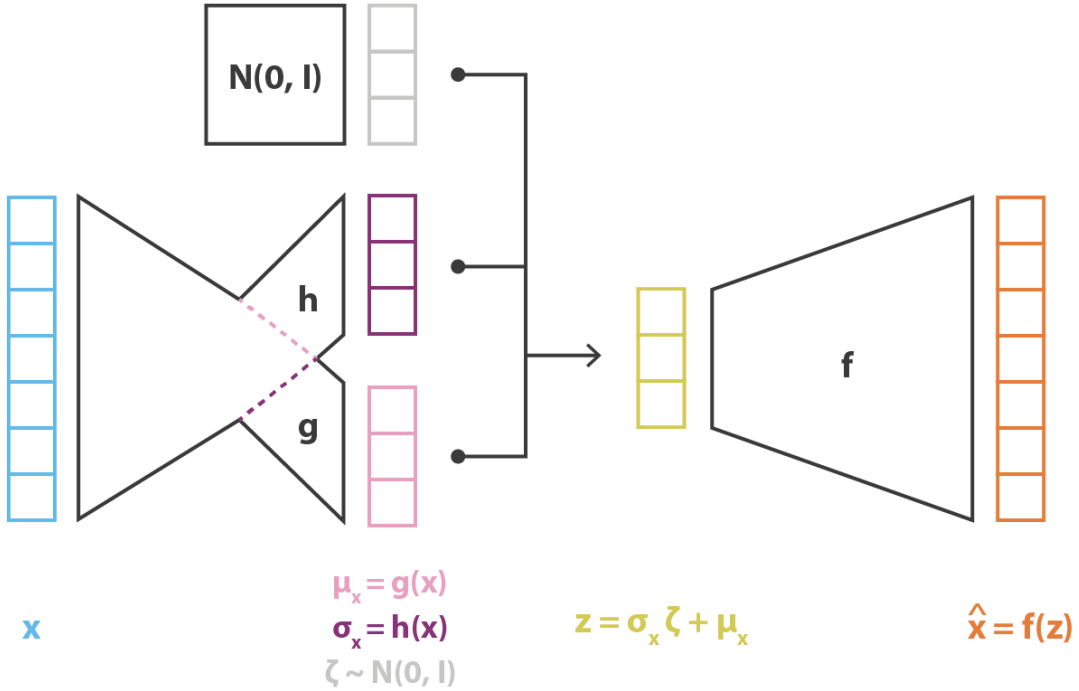
Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

KL Divergence for the normal distribution

$$
\begin{aligned}
\mathrm{KL}[P \,||\, Q] &= \frac{1}{2}\left[-\frac{\sigma_1^2}{\sigma_1^2} + \frac{\mu_1^2 + \sigma_1^2 - 2\mu_2\mu_1 + \mu_2^2}{\sigma_2^2} - \ln\frac{\sigma_1^2}{\sigma_2^2}\right] \\
&= \frac{1}{2}\left[\frac{\mu_1^2 - 2\mu_1\mu_2 + \mu_2^2}{\sigma_2^2} + \frac{\sigma_1^2}{\sigma_2^2} - \ln\frac{\sigma_1^2}{\sigma_2^2} - 1\right] \\
&= \frac{1}{2}\left[\frac{(\mu_1 - \mu_2)^2}{\sigma_2^2} + \frac{\sigma_1^2}{\sigma_2^2} - \ln\frac{\sigma_1^2}{\sigma_2^2} - 1\right]
\end{aligned}
$$

```
kl_div = 0.5 * torch.sum(mu.pow(2) + log_var.exp() - log_var - 1)
```

Variational Auto-encoders (VAEs)



$$\text{loss} = C \, || x - \hat{x} ||^2 + KL[\, N(\mu_x, \sigma_x), N(0, I) \,] = C \, || x - f(z) ||^2 + KL[\, N(g(x), h(x)), N(0, I) \,]$$

Understanding Variational Autoencoders (VAEs), J. Rocca, 2019

VAEs for image generation

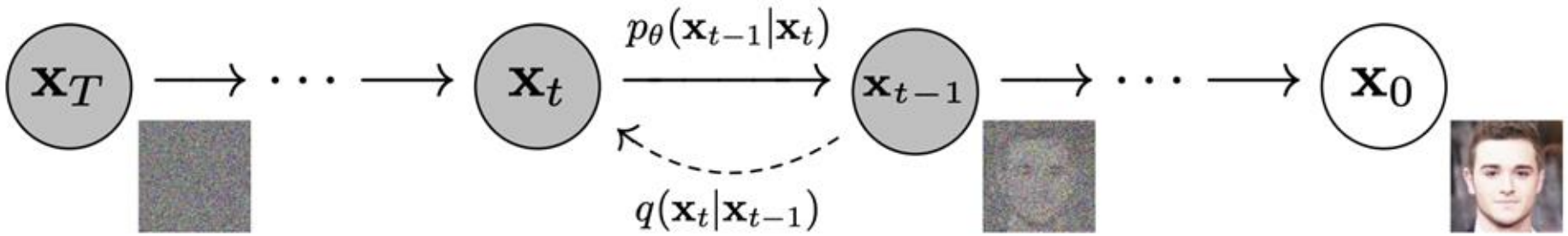
Figure 1: Class-conditional 256x256 image samples from a two-level model trained on ImageNet.

Generating Diverse High-Fidelity Images with VQ-VAE-2, Ali Razavi, Aaron van der Oord, Oriol Vinyals, 2019

Agenda for today

1. VAEs
2. Diffusion

The principle of diffusion



Denoising Diffusion Probabilistic Models (DDPM), Jonathan Ho, Ajay Jain and Pieter Abbeel, NIPS 2020

Schedulers

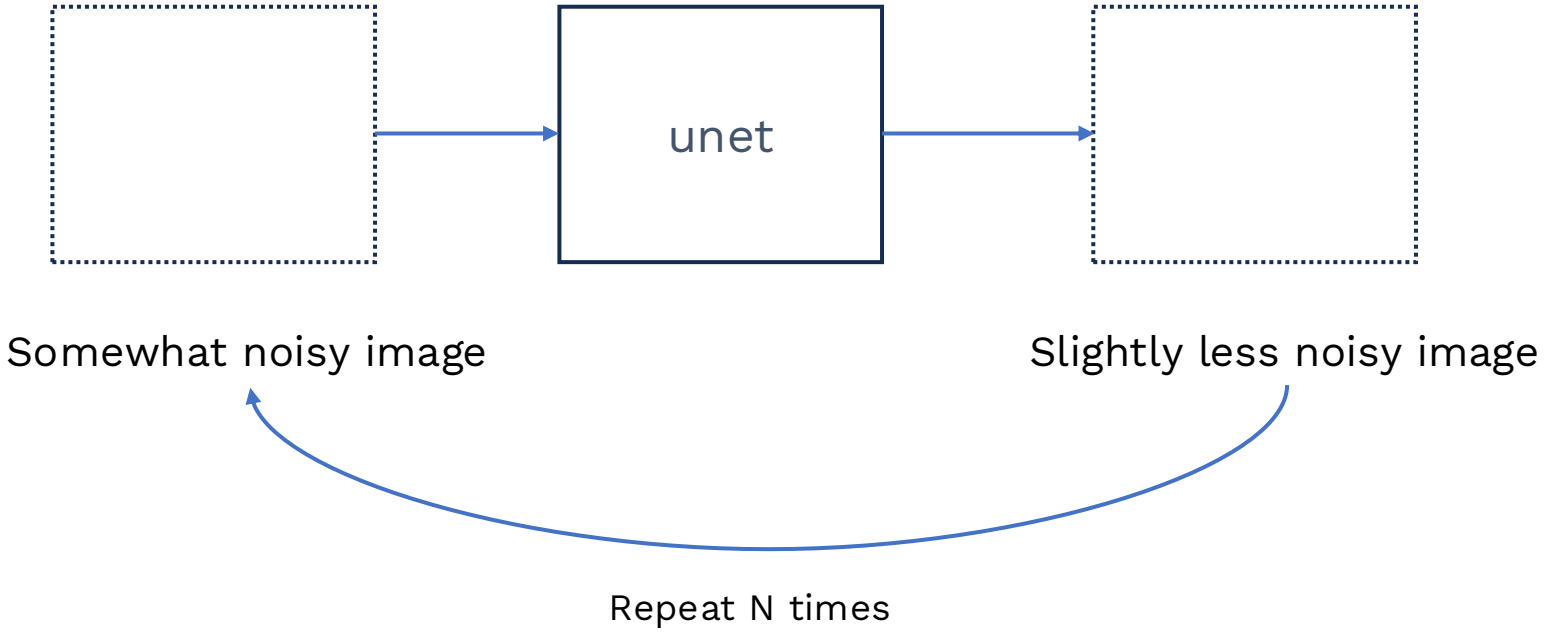
Schedulers define the methodology for iteratively adding noise to an image or for updating a sample based on model outputs.

- How to add noise for training
- How to update a sample based on an output from a pretrained model for inference

Schedulers are often defined by a *noise schedule* and an *update rule* to solve the differential equation solution.

Improved Denoising Diffusion Probabilistic Models, Nichol and Dhariwal, 2021

Stable Diffusion



U-Net

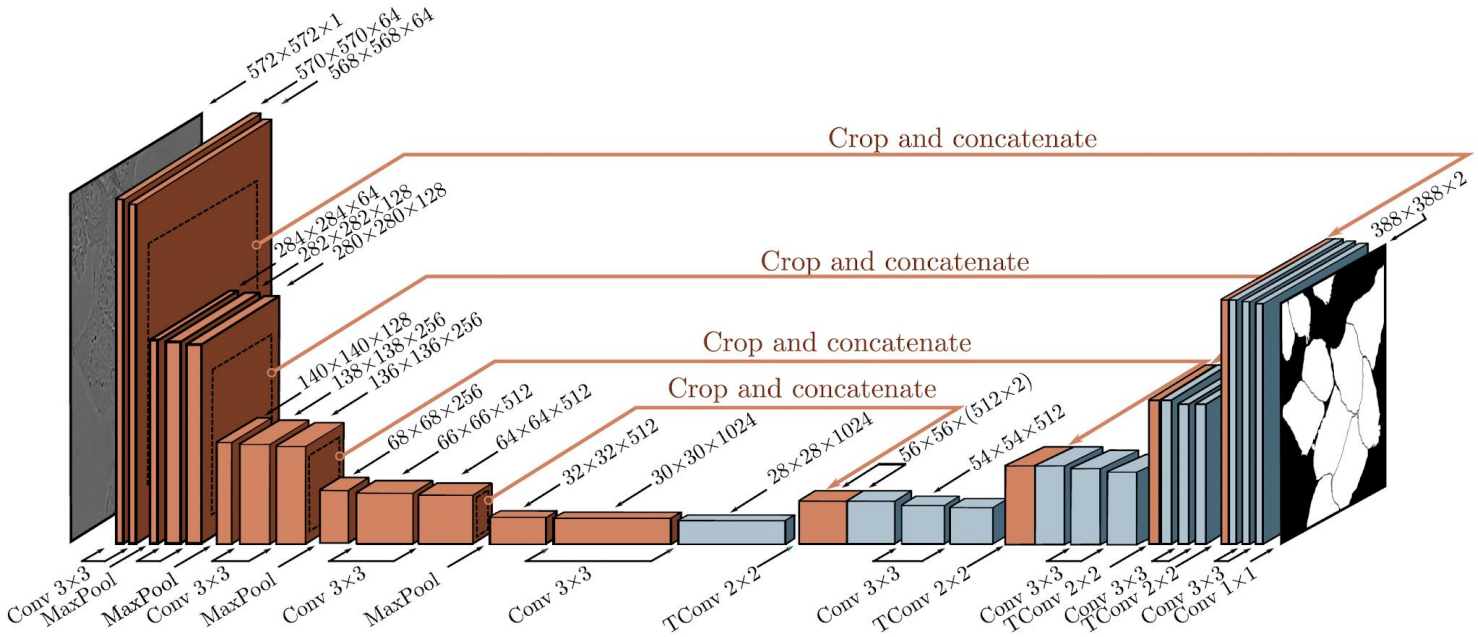
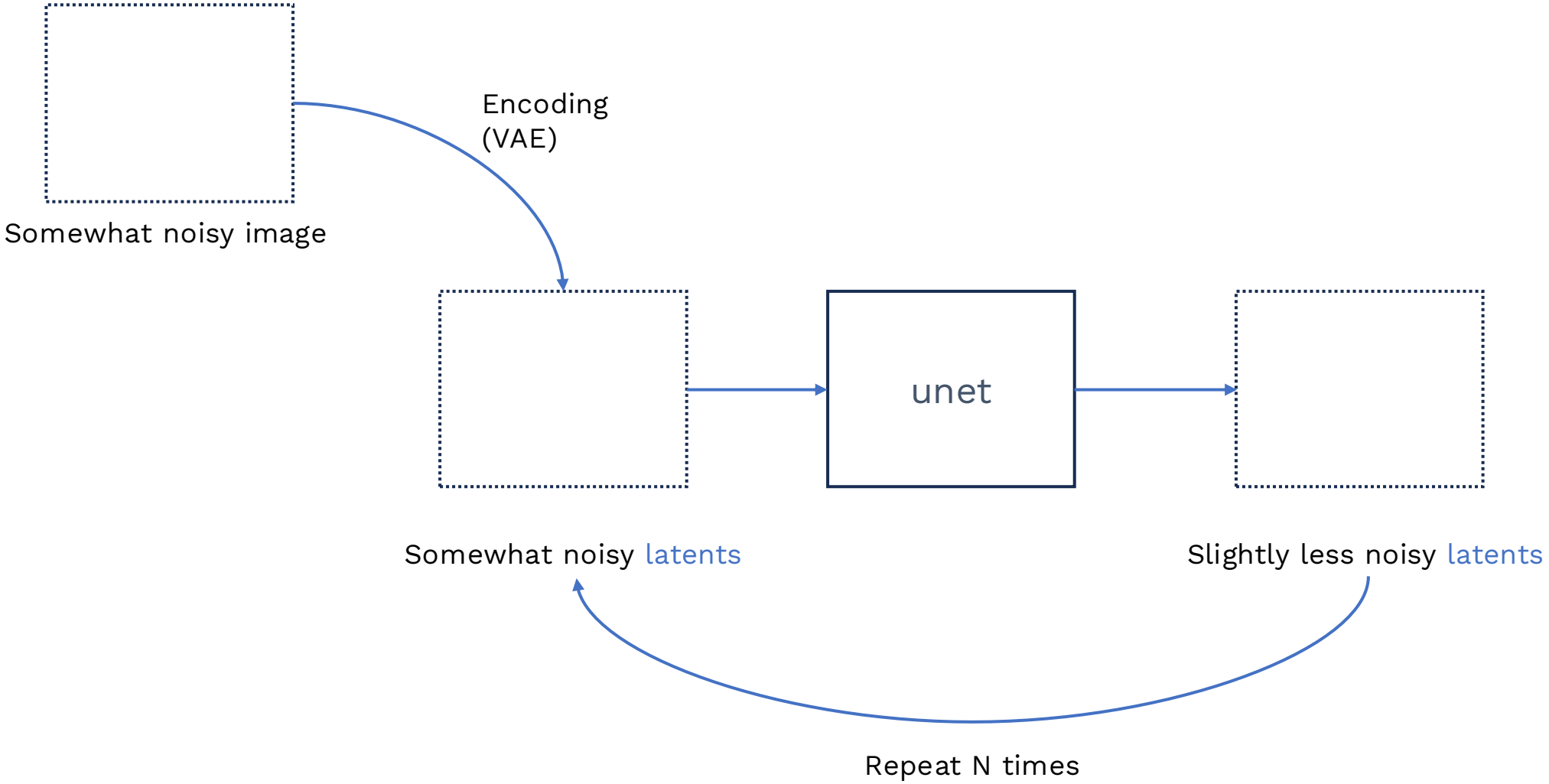


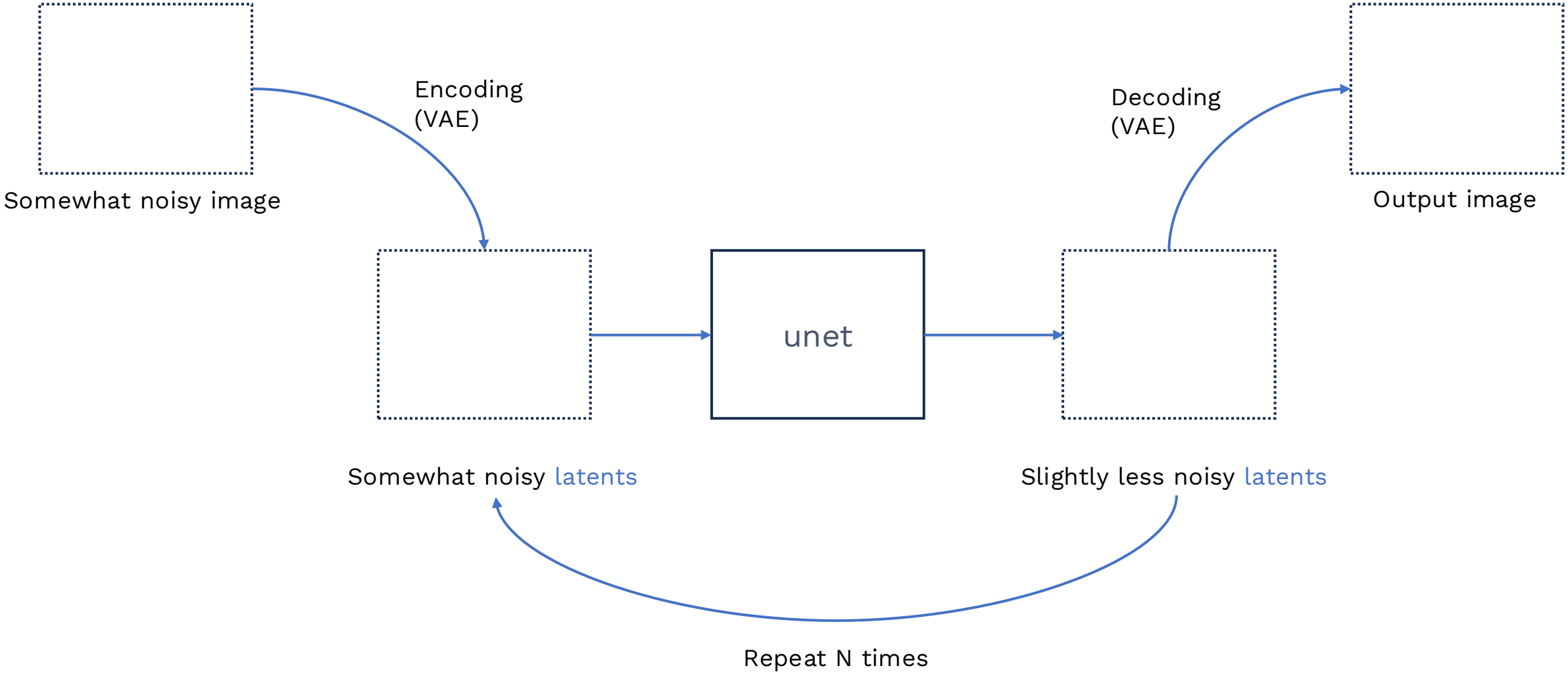
Figure 11.10 U-Net for segmenting HeLa cells. The U-Net has an encoder-decoder structure, in which the representation is downsampled (orange blocks) and then re-upsampled (blue blocks). The encoder uses regular convolutions, and the decoder uses transposed convolutions. Residual connections append the last representation at each scale in the encoder to the first representation at the same scale in the decoder (orange arrows). The original U-Net used "valid" convolutions, so the size decreased slightly with each layer, even without downsampling. Hence, the representations from the encoder were cropped (dashed squares) before appending to the decoder. Adapted from Ronneberger et al. (2015).

Source: Understanding Deep Learning, S. Prince, 2023

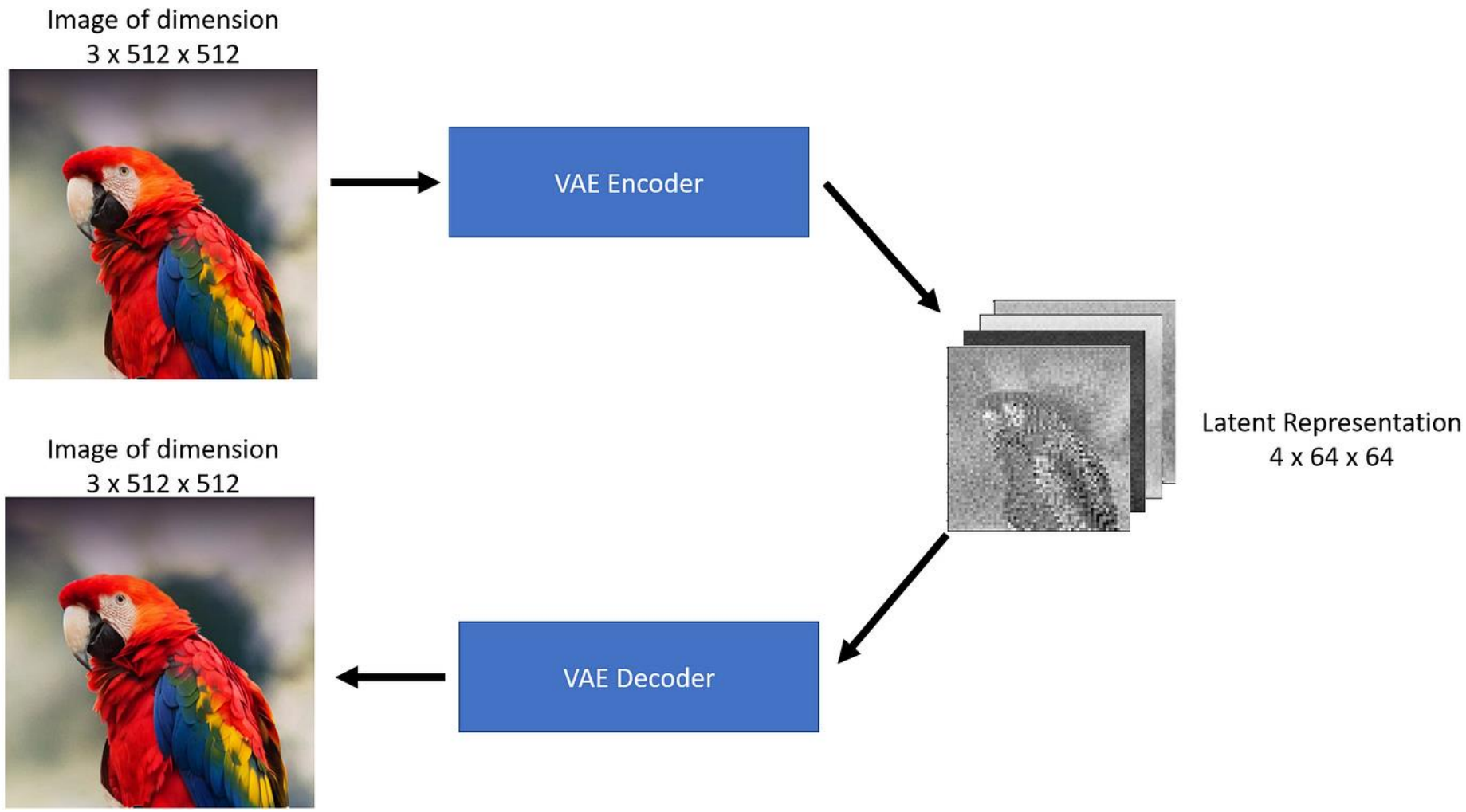
Stable Diffusion



Stable Diffusion

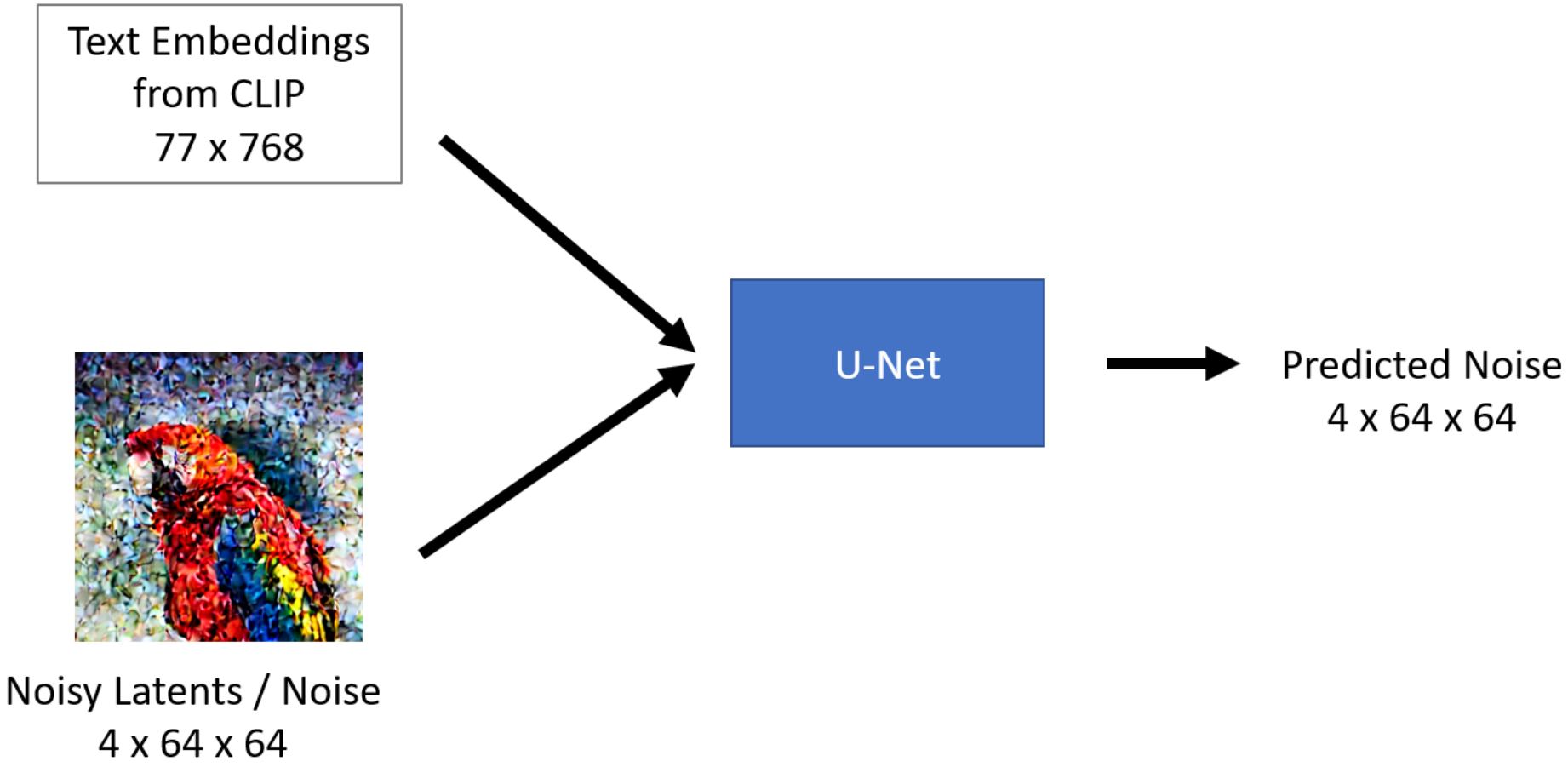


VAE Encoder



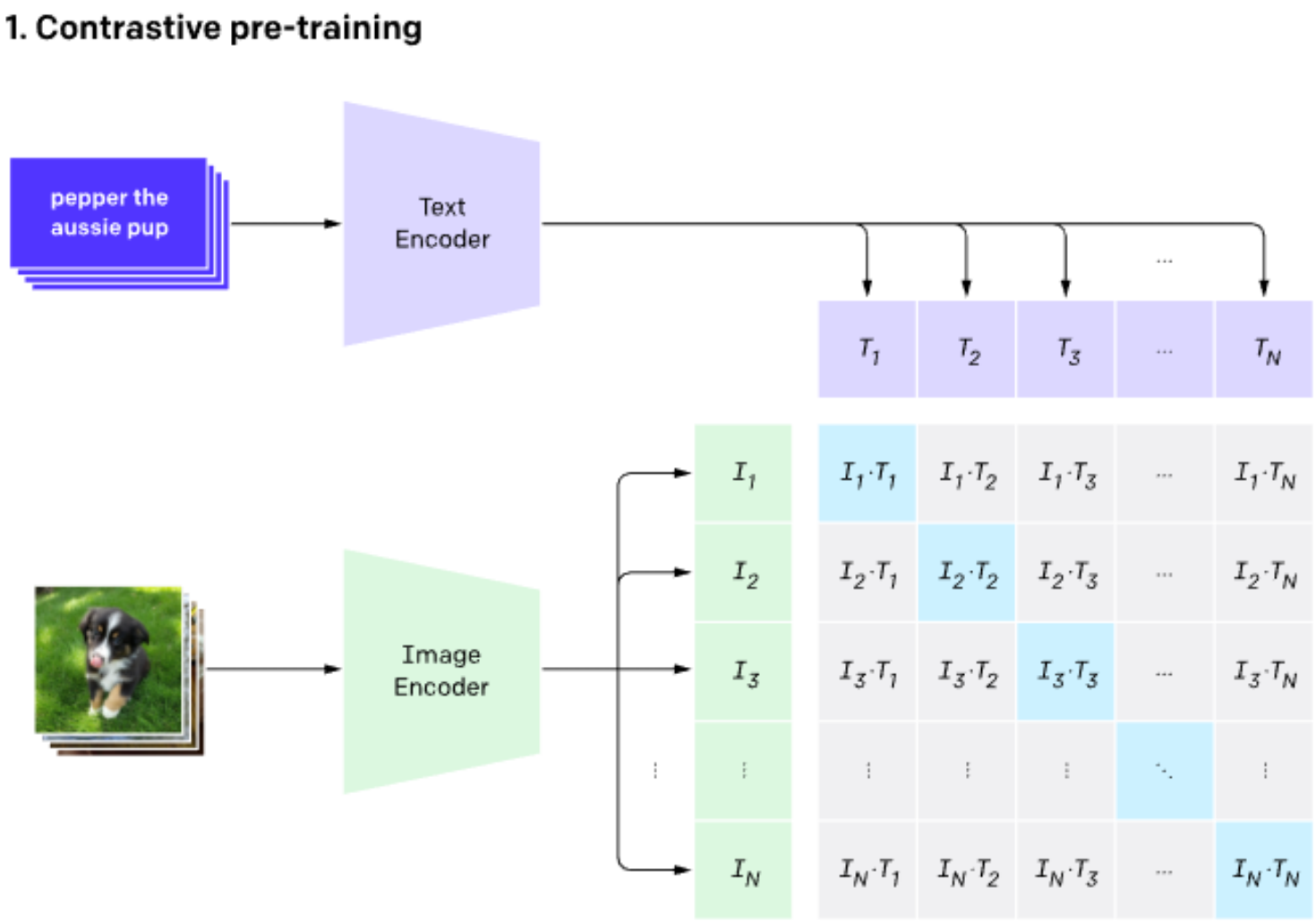
Original bird pic credit.

Guiding the diffusion with CLIP text encoding



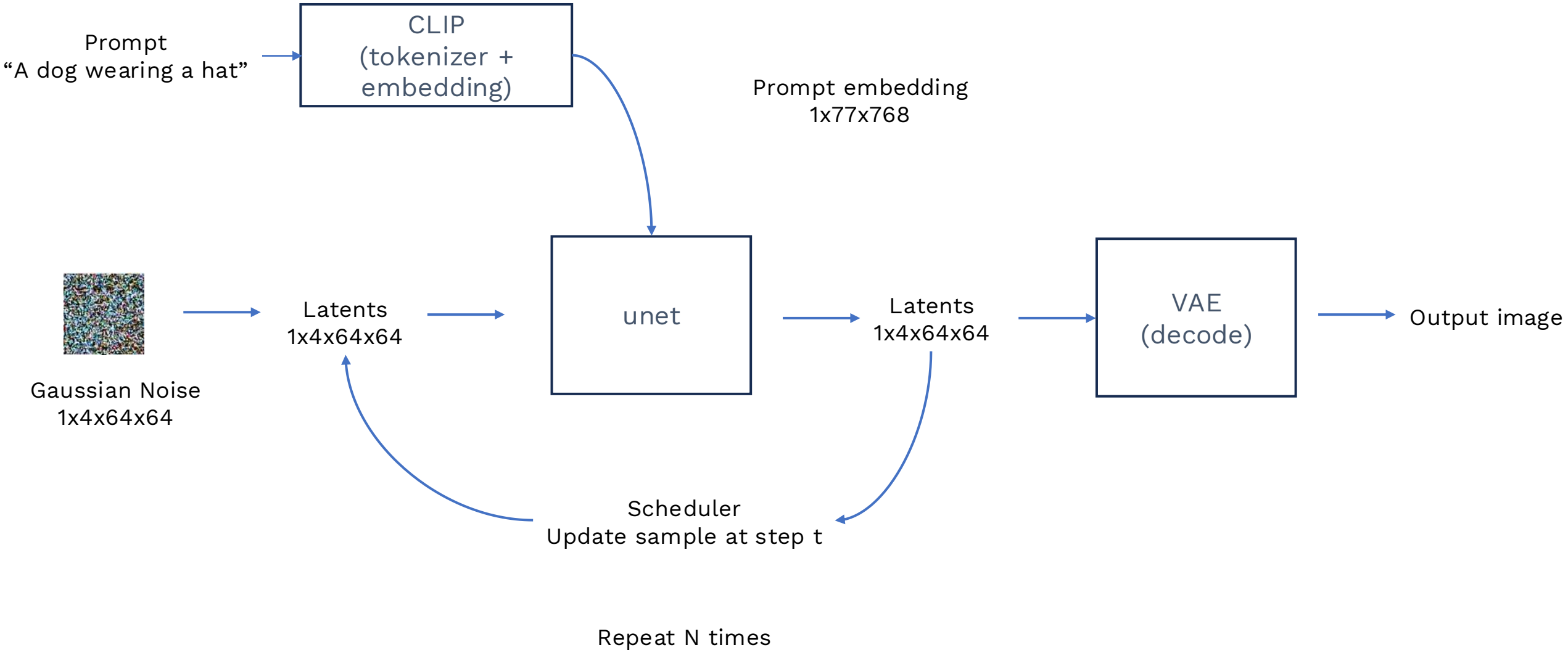
Original bird pic credit.

CLIP (Contrastive Language-Image Pretraining)

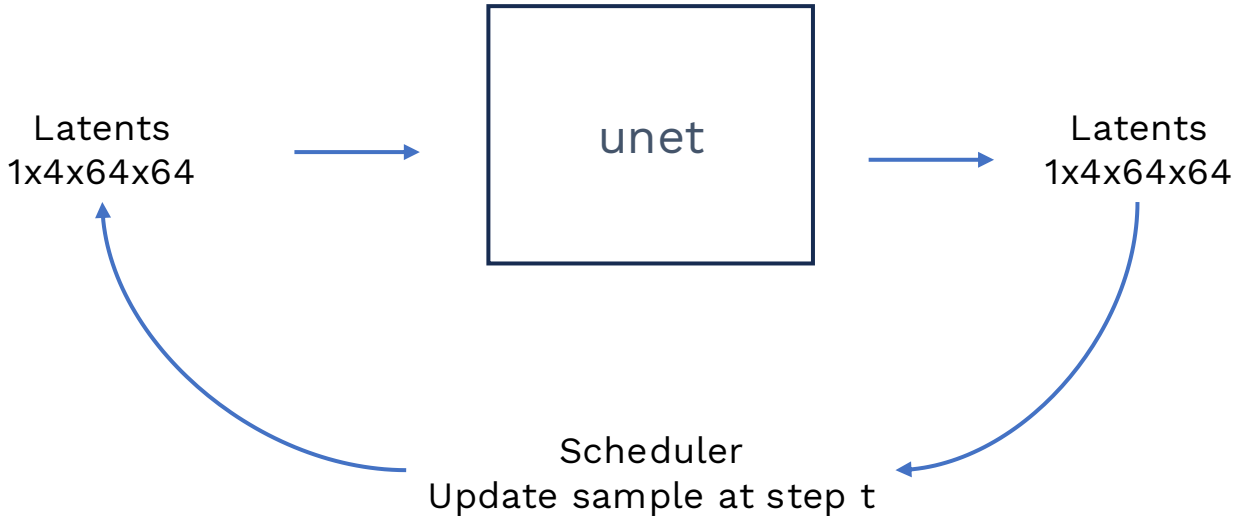


Credit — OpenAI

Stable Diffusion - Putting it all together



Common misconception



The U-Net does not predict noise between step t-1 and step t
The U-Net predicts the *entire* noise
The scheduler takes care of removing part of the noise
See demo

Going deeper

The maths of diffusion by Lilian Weng

Classifier-free guidance (CFG)

Denoising Diffusion Implicit Models (DDIM)

Diffusion Models for Text Generation (AR-Diffusion)

Diffusion Models in Reinforcement Learning (DDPO)

Diffusion Transformers (DiT)

Practical 6: let’s play with VAEs and diffusion!